

练习题

第 8 章 CMOS 逻辑电路的高级技术

1. 你的一个同学决定采用镜像电路来实现下图真值表所描述的两个输入逻辑函数。
 - a) 这一函数是否具有构成一个镜像电路所需要的正确的对称性?如果具有, 构成这个逻辑门。
 - b) 对这一情形, 镜像电路是否是一个聪明的设计?请解释。

a	b	f
0	0	0
0	1	1
1	0	0
1	1	1

图 P8.1

2. 两个串联的 PMOS 有一个公共的电容 48fF, 如图 P8.2 所示。晶体管的增益因子 $k_p = 250\mu\text{A}/\text{V}^2$ 及 $(V_{DD} - |V_{TP}|) = 2.65\text{V}$ 。用这两个晶体管同时设计一个标准的 AOI XOR 电路及一个镜像类型的 XOR 电路, 设在输出节点处总的输出电容为 $C_{out} = 175\text{fF}$, 求这两个设计的 t_{pLH} 值。

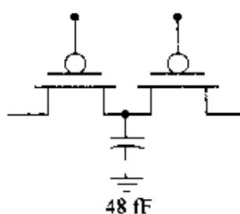


图 P8.2

3. 用 $0.25\mu\text{m}$ CMOS 工艺 (用讲义表 4.2 的参数), 电源电压 $V_{DD} = 2.5\text{V}$, 设计一个伪 nMOS NAND2 门和一个伪 nMOS NAND3 门, 两个门都要求 $V_{OL} = 0.3\text{V}$, PMOS 的宽长比为 2。比较这两个门的晶体管尺寸。
4. 画出能实现如下逻辑操作的伪 nMOS 电路:

$$(a) f = \overline{a \cdot b + c}; \quad (b) h = \overline{(a + b + c) \cdot x + y \cdot z}; \quad (c) F = \overline{a + (c \cdot [x + (y \cdot z)])}$$

5. 考虑如下两个对耦表达式:

$$g = \overline{x \cdot y + z \cdot w} \quad G = \overline{(x + y) \cdot (z + w)}$$

当采用伪 nMOS 设计时哪种形式 (AOI 或 OAI) 可以提供最佳性能?

6. 设计一个三态电路, 它在控制信号 $T=1$ 时处于高阻状态, 而在 $T=0$ 时其作用如一个非反相的缓冲器。

7. 画出具有如下输出的一个动态逻辑门的电路图:

$$F = \overline{a \cdot (b + c + d)}$$

8. 画出具有如下输出的一个动态逻辑门的电路图, 采用最少数目的晶体管:

$$f = \overline{a \cdot b + c \cdot a}$$

9. 通过建立 NMOS 逻辑树结构, 就图 P8.3 所示功能表设计一个 DCVSL 逻辑门。

f	1	1	0	1	0	0	1	1
c	0	1	0	1	0	1	0	1
b	0	0	1	1	0	0	1	1
a	0	0	0	0	1	1	1	1

图 P8.3

10. Compute the following for the pseudo-NMOS inverter shown in Figure P8.4:

- V_{OL} and V_{OH}
- NM_L and NM_H
- The power dissipation: (1) for V_{in} low, and (2) for V_{in} high
- For an output load of 1pF, calculate t_{pLH} , t_{pHL} , and t_p . Are the rising and falling delays equal? Why or why not?

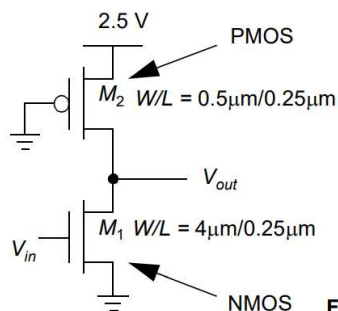


图 P8.4

11. Implement $F = (ABC)' + A'CD$ (and F') in DCVSL. Assume A , B , C , D , and their complements are available as inputs. Use the minimum number of transistors.

12. A complex logic gate is shown in Figure P8.5.

- Write the Boolean equations for outputs F and G . What function does this circuit implement?
- What logic family does this circuit belong to?

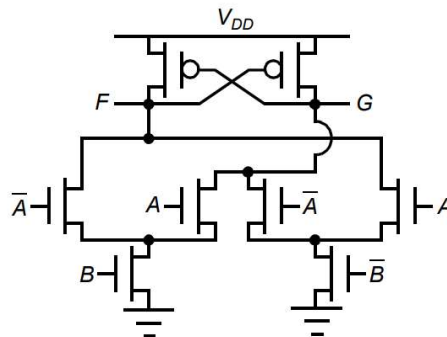


图 P8.5

13. Implement the function $S = ABC + AB'C' + A'B'C + A'BC'$, which gives the sum of two inputs with a carry bit, using NMOS pass transistor logic. Design a DCVSL gate which implements the same function. Assume A , B , C , and their complements are available as inputs.

14. Figure P8.6 contains a pass-gate logic network.

- Determine the truth table for the circuit. What logic function does it implement?
- Assuming 0 and 2.5 V inputs, size the PMOS transistor to achieve a $V_{OL} = 0.3$ V.
- If the PMOS were removed, would the circuit still function correctly? Does the PMOS transistor serve any useful purpose?

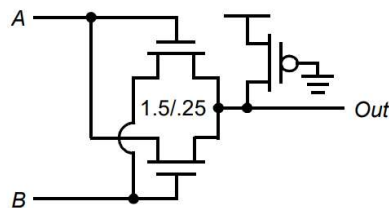


图 P8.6 Pass-gate network

15. Consider the circuit of Figure P8.7. Let $C_x = 50$ fF, M_r has $W/L = 0.375\mu\text{m} / 0.375\mu\text{m}$, M_n has $W/L_{\text{eff}} = 0.375\mu\text{m} / 0.25\mu\text{m}$. Assume the output inverter doesn't switch until its input equals $V_{DD}/2$.

- How long will it take M_n to pull down node x from 2.5V to 1.25V if I_n is at

- 0V and B is at 2.5V?
- How long will it take M_n to pull up node x from 0V to 1.25V if V_{in} is 2.5V and V_B is 2.5V?
 - What is the minimum value of V_B necessary to pull down V_x to 1.25V when $V_{in} = 0V$?

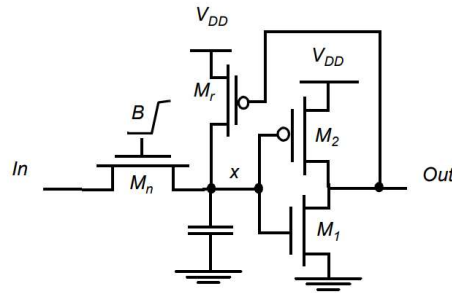


图 P8.7 Level restorer

- Sketch the waveforms at x , y , and z for the given inputs (Figure P8.8). You may approximate the time scale, but be sure to compute the voltage levels. Assume that $V_T = 0.5V$ when body effect is a factor.

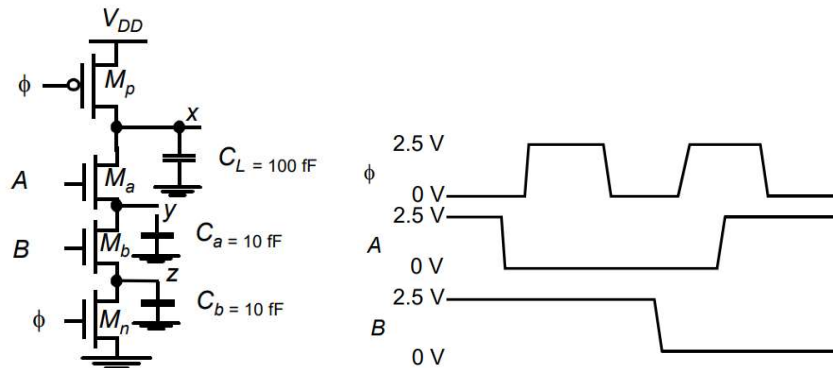


图 P8.8

- Pass Transistor Logic. Consider the circuit of Figure P8.9. Assume the inverter switches ideally at $V_{DD}/2$, neglect body effect, channel length modulation and all parasitic capacitance throughout this problem.
 - What is the logic function performed by this circuit?
 - Explain why this circuit has non-zero static dissipation.
 - Using only just 1 transistor, design a fix so that there will not be any static power dissipation. Explain how you chose the size of the transistor.
 - Implement the same circuit using transmission gates.
 - Replace the pass-transistor network in Figure P8.9 with a pass transistor network that computes the following function: $x = ABC$ at the node x .

Assume you have the true and complementary versions of the three inputs A , B and C .

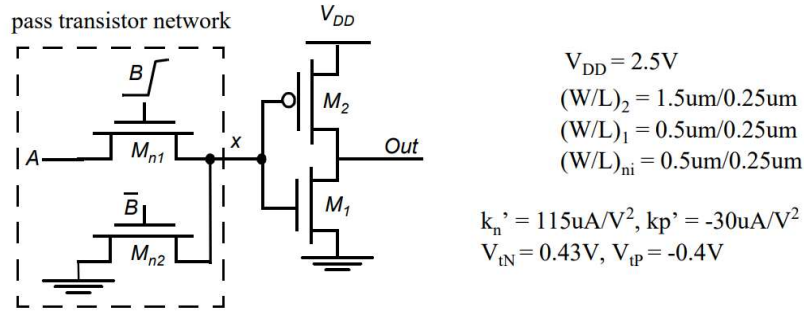


图 P8.9

18. Figure P8.10 shows a dynamic CMOS circuit in Domino logic.

- What Boolean functions are implemented at outputs F and G ? If A and B are interpreted as two-bit binary words, $A = A_1A_0$ and $B = B_1B_0$, then what interpretation can be applied to output G ?
- Which gate (1 or 2) has the highest potential for harmful charge sharing and why? What sequence of inputs (spanning two clock cycles) results in the worst-case charge-sharing scenario?
- If you determined which gate (1 or 2) suffers the most from charge sharing. Add a single $2\mu m/0.25\mu m$ PMOS precharge transistor (with its gate driven by the clock ϕ and its source connected to V_{DD}) to one of the nodes in that gate to maximally reduce the chargesharing effect. What effect (if any) will this addition have on the gate delay? Use SPICE to demonstrate that the additional transistor has eliminated charge sharing for the previously determined worst-case sequence of inputs.

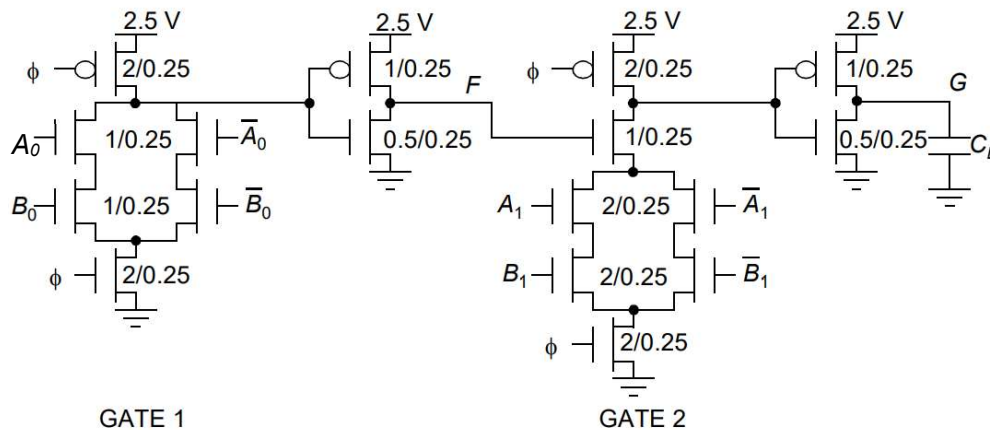


图 P8.10 DOMINO logic circuit.

19. Consider a conventional 4-stage Domino logic circuit as shown in Figure P8.11 in which all precharge and evaluate devices are clocked using a common clock ϕ . For this entire problem, assume that the pulldown network is simply a single NMOS device, so that each Domino stage consists of a dynamic inverter followed by a static inverter. Assume that the precharge time, evaluate time, and propagation delay of the static inverter are all $T/2$. Assume that the transitions are ideal (zero rise/fall times).

- Complete the timing diagram for signals *Out1*, *Out2*, *Out3* and *Out4*, when the *IN* signal goes high before the rising edge of the clock ϕ . Assume that the clock period is $10T$ time units.
- Suppose that there are no evaluate switches at the 3 latter stages. Assume that the clock ϕ is initially in the precharge state ($\phi=0$ with all nodes settled to the correct precharge states), and the block enters the evaluate period ($\phi=1$). Is there a problem during the evaluate period, or is there a benefit? Explain.
- Assume that the clock ϕ is initially in the evaluate state ($\phi=1$), and the block enters the precharge state ($\phi=0$). Is there a problem, or is there any benefit, if the last three evaluate switches are removed? Explain.

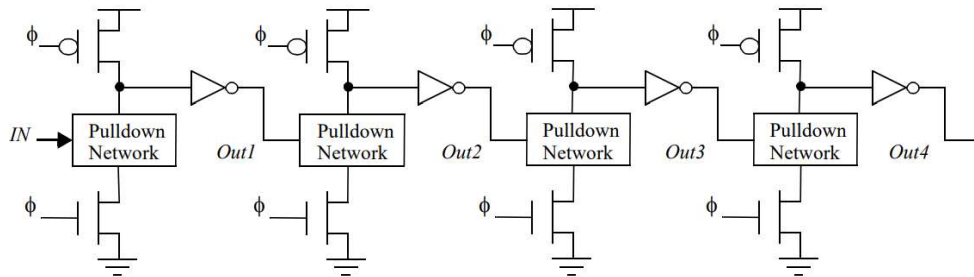


图 P8.11 Conventional DOMINO Dynamic Logic.